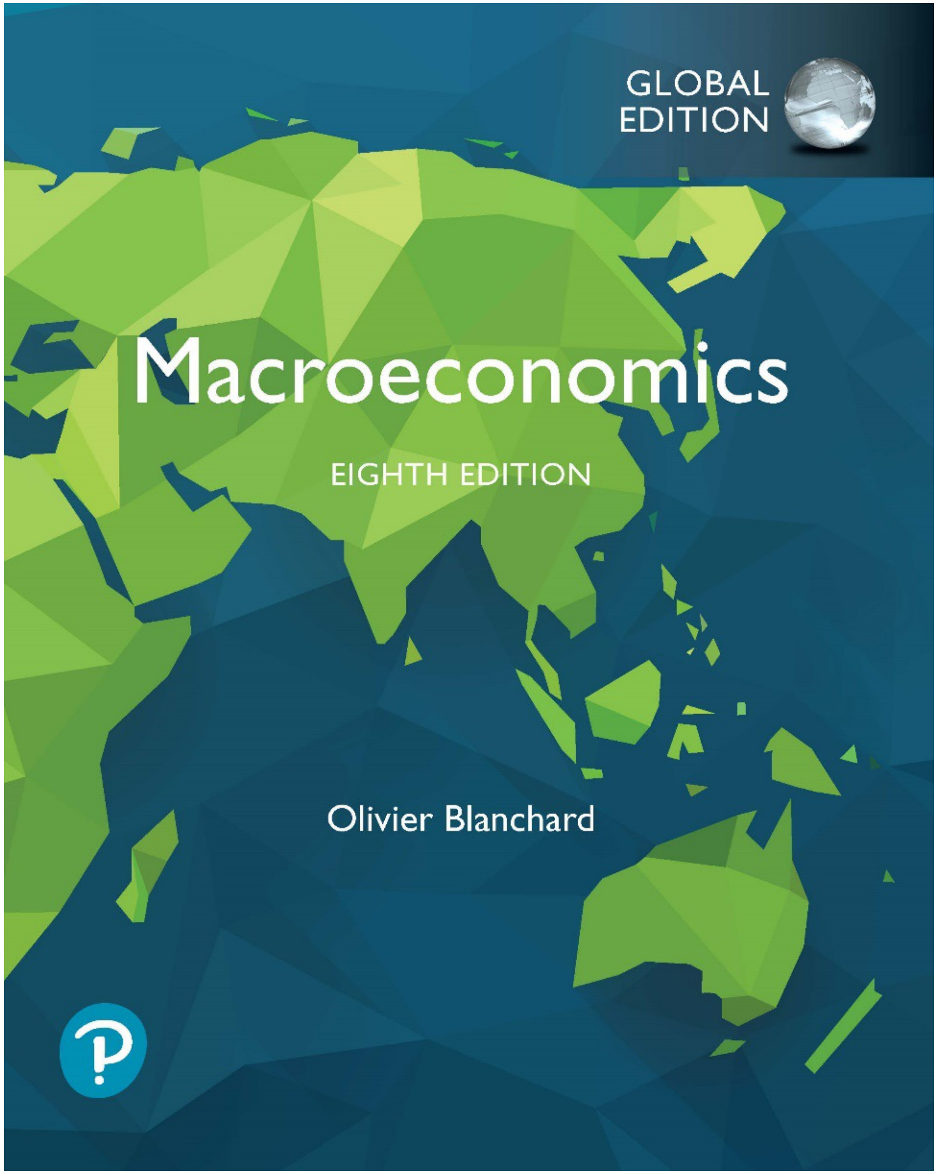
Macroeconomics

Eighth Edition, Global Edition



Chapter 15

Expectations, Consumption, and Investment

Chapter 15 Outline

Expectations, Consumption, and Investment

15.1 Consumption

15.2 Investment

15.3 The Volatility of Consumption and Investment

APPENDIX Derivation of the Expected Present Value of Profits under Static Expectations

Expectations, Consumption, and Investment

- In this chapter, we look at the role expectations play in determining the two main components—consumption and investment.
- This description of consumption and investment will be the main building block of the expanded *IS-LM* model we develop in Chapter 16.

15.1 Consumption (1 of 4)

- The modern theory of consumption was developed independently in the 1950s by Milton Friedman as the **permanent theory of consumption**, and by Franco Modigliani as the **life cycle theory of consumption**.
- Consumption for a *very foresighted consumer* depends on:
  - **Financial wealth**: The value of checking and saving accounts
  - **Housing wealth**: The value of the house owned minus the mortgage due
  - **Human wealth**: After-tax labor income over working life
  - **Nonhuman wealth**: The sum of financial wealth and housing wealth

FOCUS: Up Close and Personal: Learning from Panel Data Sets

- **Panel data sets** are data sets that show the value of one or more variables for many individuals or many firms over time.
- One example is the *Panel Study of Income Dynamics* (*PSID*), which started in 1968 with approximately 4,800 families.
- Each year, the survey asks people about their income, wage rate, number of hours worked, health, and food consumption.

15.1 Consumption (2 of 4)

- Consumption decision:

$$C_t = C(\text{total wealth}_t) \qquad (15.1)$$

- If you want to consume the same amount every year, the constant level of consumption that you can afford equals your total wealth divided by your expected remaining life.
- However, you might:
  - not want to plan for constant consumption over your lifetime
  - make consumption decisions in a less forward-looking fashion
  - become unemployed or sick
  - not be able to get a loan from a bank when you need to borrow

Focus: Do People Save Enough for Retirement?

Table 1 Mean Wealth of People, Age 65–69, in 2008 (in thousands of 2008 dollars)

	Married couples	Single person household
Social Security pension	262	134
Employer-provided pension	129	63
Personal retirement assets	182	47
Other financial assets	173	83
Home equity	340	188
Other equity	69	18
Total	1,155	533

- *Source:* James M. Poterba, Steven F. Venti, and David A. Wise, “The composition and drawdown of wealth in retirement,” *Journal of Economic Perspectives*, 25(4), pages 95–118, Fall 2011.

15.1 Consumption (3 of 4)

- Consumption function with after-tax labor income:

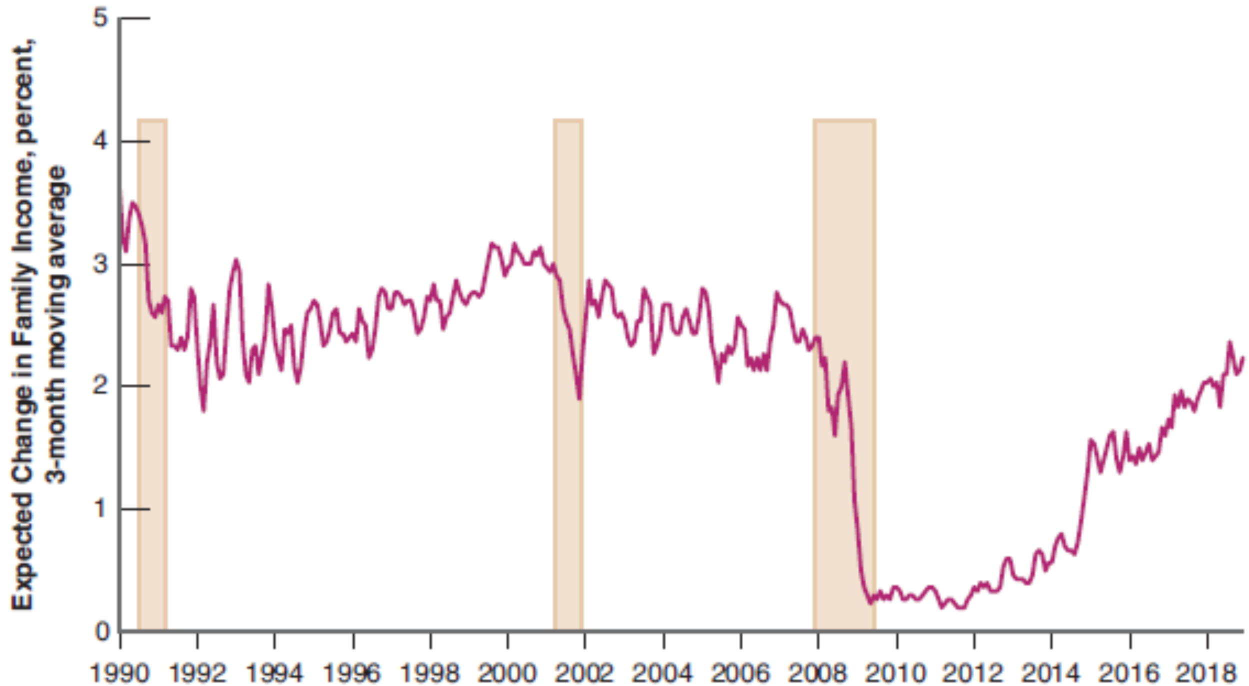
$$C_t = \underset{(+,+)}{C(\text{total wealth}_t, Y_{Lt} - T_t)} \qquad (15.2)$$

- Expectations affect consumption:
  - Directly through human wealth
  - Indirectly through nonhuman wealth
- Implications of expectations of the relation between consumption and income:
  - *Consumption is likely to respond less than one-for-one in current income.*
  - *Consumption may move even if current income does not change.*

15.1 Consumption (4 of 4)

Figure 15.1 Expected Change in Family Income since 1990

After falling sharply in 2008 and 2009, expectations of income growth remained low for a long time.



- *Source:* Survey of Consumers, Table 14, University of Michigan, https://data.sca.isr.umich.edu.

The shaded areas represent recessions.

15.2 Investment (1 of 5)

- To compute the present value of expected profits the firm must first estimate how long the machine will last or the depreciation rate δ.
- The *present value of expected profits* from buying the machine in year *t* is:

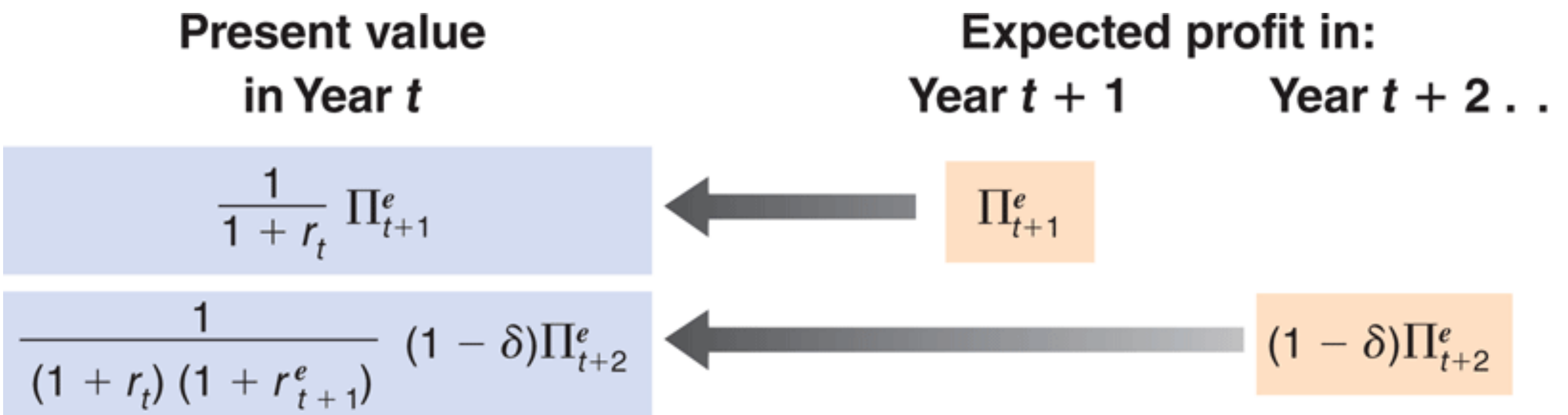
$$V(\Pi_t^e) = \frac{1}{1+r_t}\Pi_{t+1}^e + \frac{1}{(1+r_t)(1+r_{t+1}^e)}(1-\delta)\Pi_{t+2}^e + \dots \qquad (15.3)$$

- The investment function becomes:

$$I_t = \underset{(+)}{I}[V(\Pi_t^e)] \qquad (15.4)$$

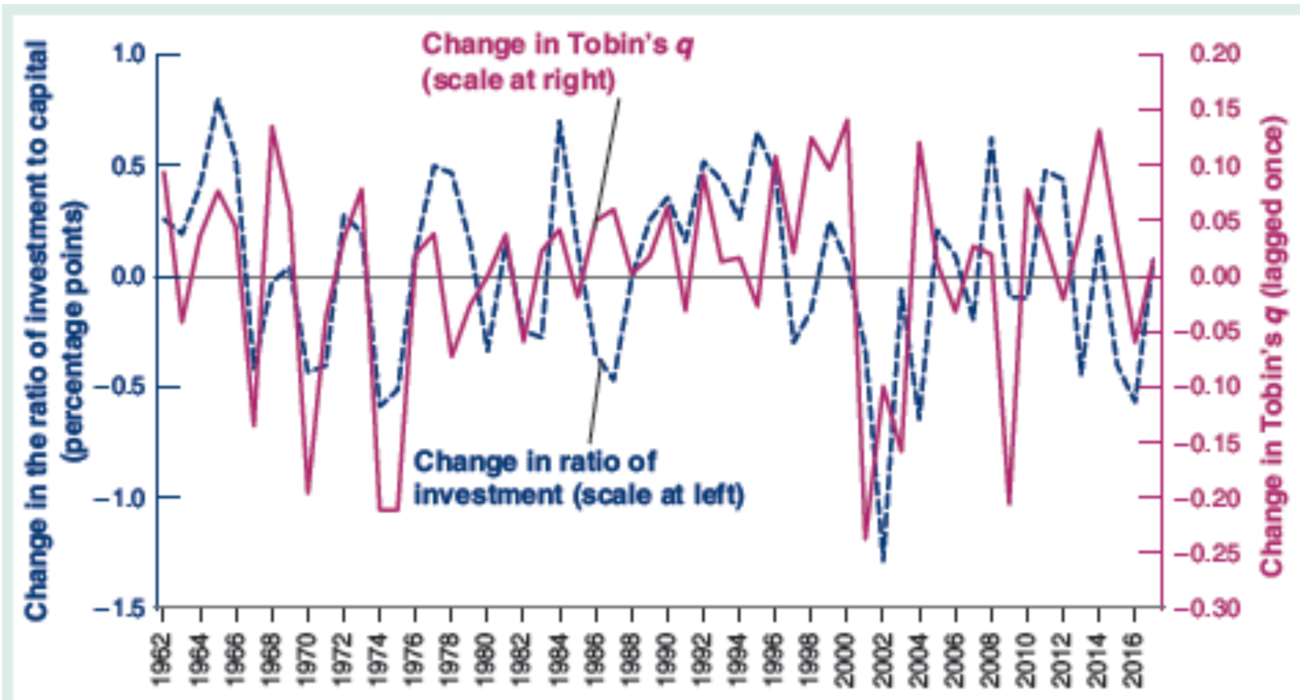
15.2 Investment (2 of 5)

Figure 15.2 Computing the Present Value of Expected Profits



FOCUS: Investment and the Stock Market

- **Figure 1** Tobin's *q* versus the Ratio of Investment to Capital. Annual Rates of Change since 1962



- *Source:* Flow of Funds, Table s5a. Numerator of *q*: Market value of equity + Debt + Loans – Financial Assets, nonfinancial US corporations. Denominator: Nonfinancial assets of nonfinancial US corporations..

15.2 Investment (3 of 5)

- Assume **static expectations**:

$$\Pi^e_{t+1} = \Pi^e_{t+2} = \ldots = \Pi_t$$
$$r^e_{t+1} = r^e_{t+2} = \ldots = r_t$$

so that equation (15.3) becomes:

$$V(\Pi_t) = \frac{\Pi_t}{r_t + \delta} \qquad (15.5)$$

- Replacing equation (15.5) in equation (15.4) gives:

$$I_t = I\left(\frac{\Pi_t}{r_t + \delta}\right) \qquad (15.6)$$

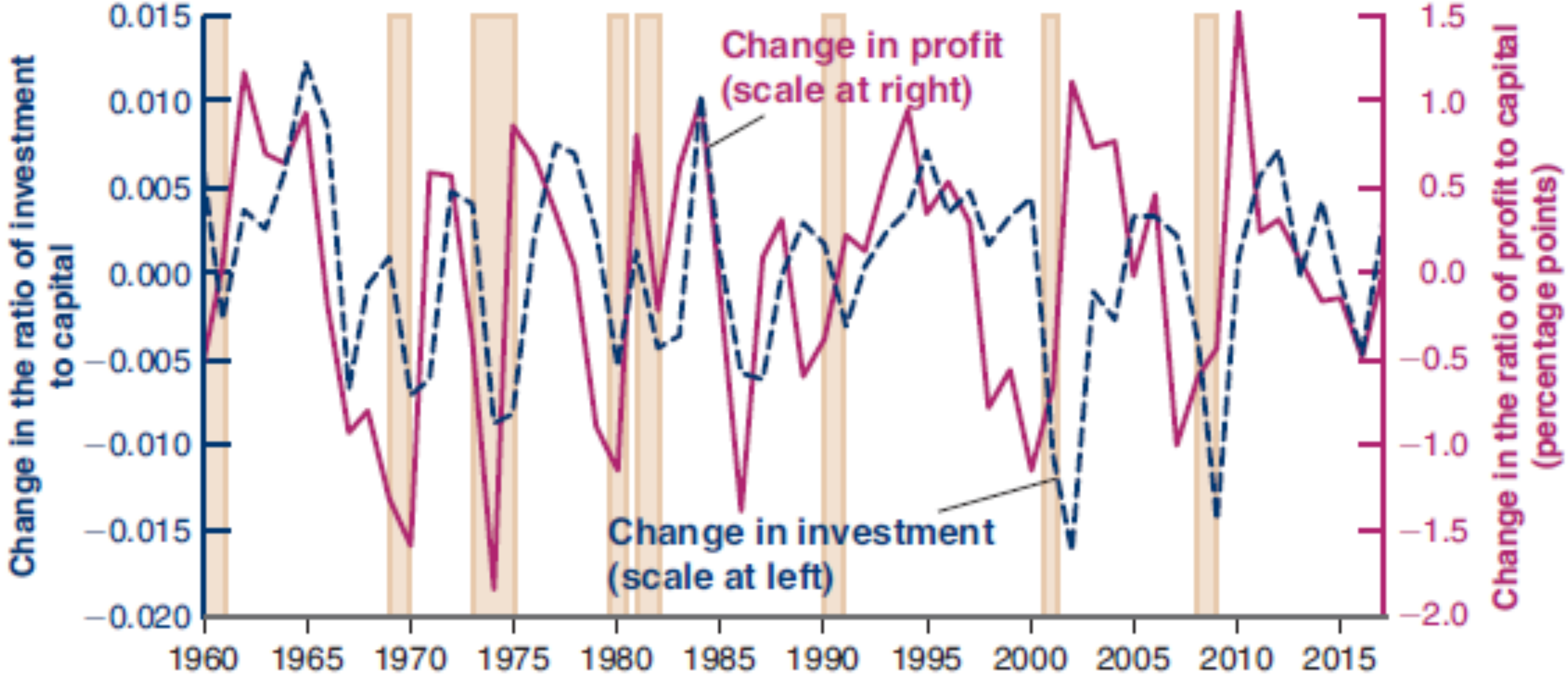
where $(r_t + \delta)$ is rental cost (user cost), or shadow cost of capital.

15.2 Investment (4 of 5)

- According to the investment function of equation (15.6):
  - *Investment depends on the ratio of profit to the user cost.*
  - *The higher the profit, the higher the level of investment.*
  - *The higher the user cost, the lower the level of investment.*
- If a firm's current profit is low:
  - it can get the funds it needs only by borrowing if it wants to buy new machines.
  - it might have difficulty borrowing even if it wants to invest.

15.2 Investment (5 of 5)

Figure 15.3 Changes in Investment and Profit in the United States since 1960



- *Source:* Gross investment: Federal Reserve Board, Flow of Funds, series FA105013005.A. Capital stock: BEA Fixed Assets Tables, net stock of private nonresidential fixed assets, nonfinancial. Profit: BEA, NIPA Table 1.14, Net operating surplus minus taxes, minus transfers, minus net interest payments.

FOCUS: Profitability versus Cash Flow

- **Profitability** is the expected present discount value of future profits.
- **Cash flow** is net flow of cash the firm receives now (current profit).
- An economist found that in 1986 when the declines in the oil price led to large losses in oil-related activities, investment spending in nonoil activities also reduced.
- This suggests that current cash flow matters in investment.

15.2 Investment (1 of 2)

- Investment depends both on the expected present value of future profits and on the current level of profit:

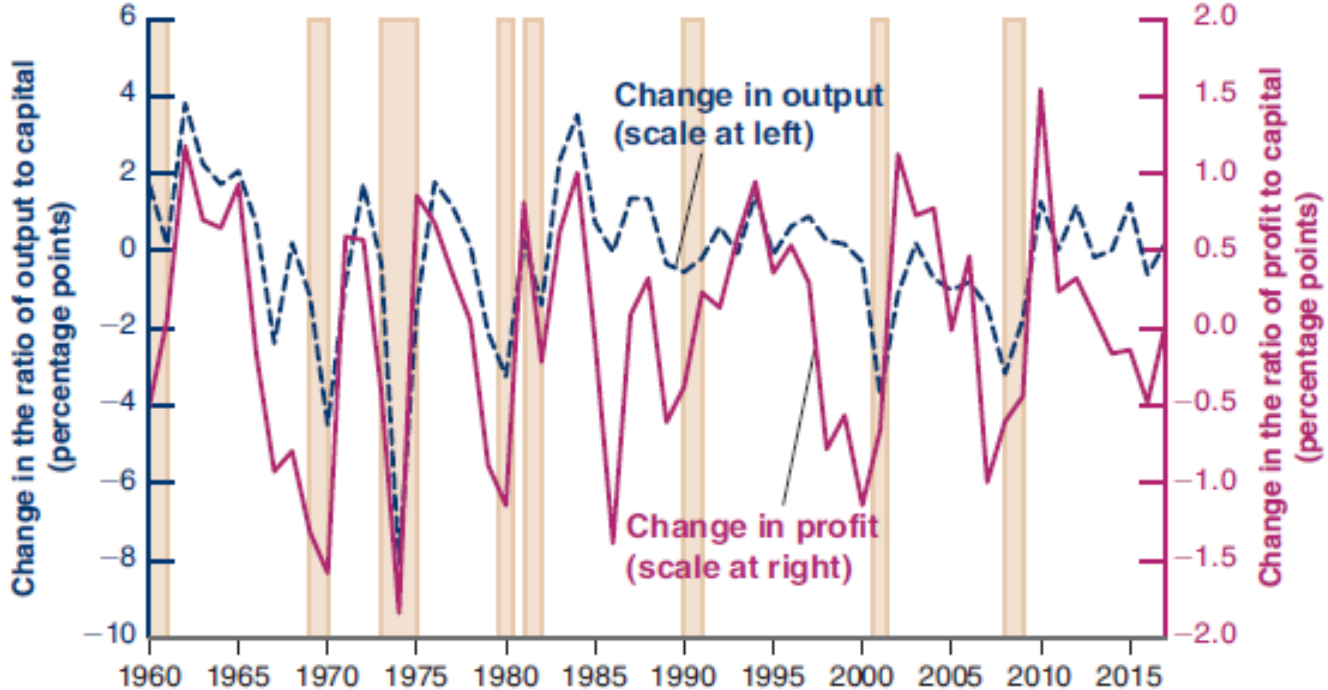
$$I_t = I[\underset{(+,+)}{V(\Pi_t^e), \Pi_t}] \qquad (15.7)$$

- Profit per unit of capital is an increasing function of the ratio of sales or output (*Y*) to the capital stock (*K*):

$$\Pi_t = \Pi\underset{(+)}{\left(\frac{Y_t}{K_t}\right)} \qquad (15.8)$$

15.2 Investment (2 of 2)

Figure 15.4 Changes in Profit per Unit of Capital versus Changes in the Ratio of Output to Capital in the United States since 1960



- *Source:* Capital stock: BEA Fixed Assets Tables. Net stock of private nonresidential fixed assets, nonfinancial assets; Profit: BEA, NIPA Table 1.14, net operating surplus minus taxes minus transfers minus net interest and miscellaneous payments; Output: BEA, Gross value added of nonfinancial corporate business sector..

15.3 The Volatility of Consumption and Investment (1 of 3)

- Similarities between consumption and investment behavior:
  - The more transitory consumers expect a current increase in income to be, the less they will increase their consumption.
  - The more transitory firms expect a current increase in sales to be, the less they revise their assessment of the present value of profits, and thus the less likely they are to buy new machines or build new factories.

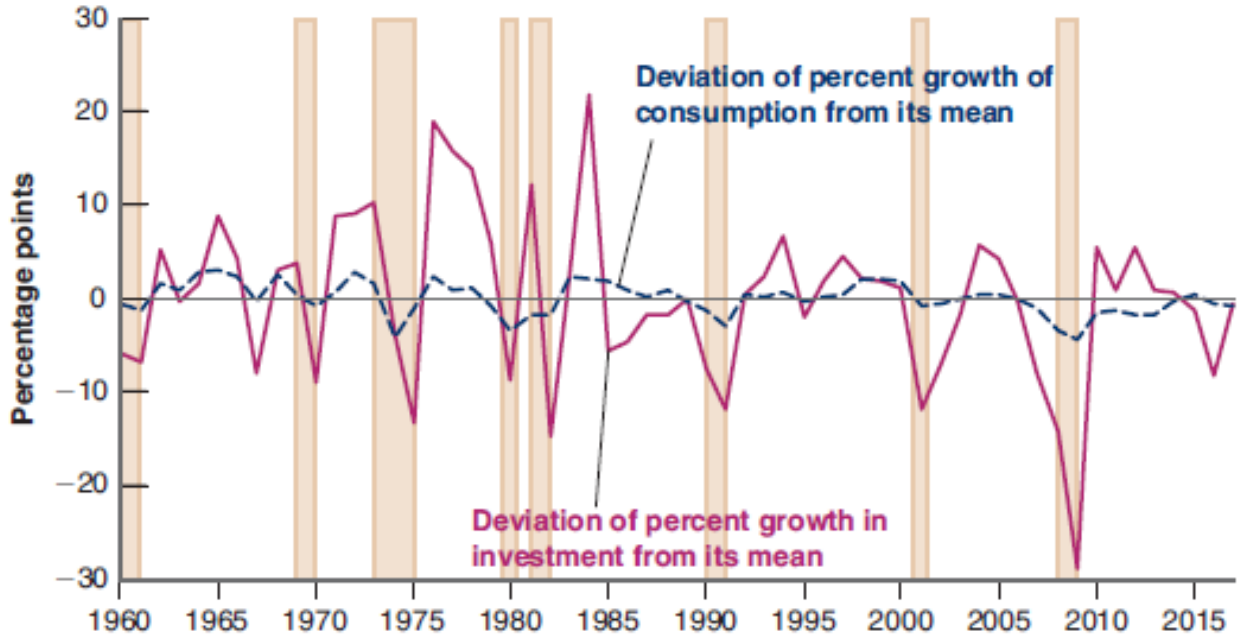
15.3 The Volatility of Consumption and Investment (2 of 3)

- Differences between consumption and investment decisions:
  - The permanent nature of the change in income implies that consumers can afford to change consumption now and in the future by the same amount as the change in income.
  - However, the change in investment spending may exceed the change in sales.

15.3 The Volatility of Consumption and Investment (3 of 3)

- Consumption and investment usually move together.
- Investment is much more volatile than consumption.
- Both components contribute roughly equal to output fluctuations.

Figure 15.5 Rates of Change in Consumption and Investment in the United States since 1960



- *Source:* FRED, series PCECC96, GPDI

APPENDIX: Derivation of the Expected Present Value of Profits under Static Expectations

- If firms expect both future profits and future interest rates to remain at the same level as today, then equation (15.3) becomes:

$$V(\Pi_t^e) = \frac{1}{1+r_t}\Pi_t + \frac{1}{(1+r_t)^2}(1-\delta)\Pi_t + \ldots$$

or

$$V(\Pi_t^e) = \frac{1}{1+r_t}\Pi_t\left(1 + \frac{1-\delta}{1+r_t} + \ldots\right) \qquad (15.\mathrm{A}1)$$

which can be reduced to:

$$V(\Pi_t^e) = \frac{1}{1+r_t}\frac{1+r_t}{r_t+\delta}\Pi_t$$

which can be further simplified to produce equation (15.5) in the text.